

X-RAY SECONDARY HEATING AND IONIZATION IN QUASAR EMISSION-LINE CLOUDS

J. MICHAEL SHULL¹ AND MICHAEL E. VAN STEENBERG¹

Joint Institute for Laboratory Astrophysics, University of Colorado and National Bureau of Standards, and
 Laboratory for Atmospheric and Space Physics

Received 1985 March 20; accepted 1985 May 7

ABSTRACT

We present accurate Monte Carlo computations of the X-ray secondary electron heating, ionization, and excitation of H and He gas in interstellar space and in quasar emission-line clouds. The fraction of energy deposited in each form is sensitive to the background ionization fraction, $x = n(\text{H}^+)/n(\text{H}_{\text{tot}})$, and can affect the temperature, ionization state, and line emissivities at large depths in X-ray photoionized clouds. We provide analytic fits for these energy fractions over the range $0.0001 < x < 1$ for primary electron energies up to many keV. In both broad-line and narrow-line clouds, emission lines sensitive to the energy budget and electron density may be strongly affected.

Subject headings: galaxies: nuclei — line formation — quasars

I. INTRODUCTION

The broad emission lines seen in the spectra of active galactic nuclei (AGN) are currently believed to arise from the reprocessing of ionizing UV/X-ray radiation by a collection of high-velocity “broad-line clouds” surrounding a central object (Osterbrock 1984). In order to derive physical information about the clouds and the ionizing source, astronomers construct photoionization models which incorporate complex atomic processes, radiative transfer, and assumptions about source flux and geometry.

The first photoionization models of QSO emission lines assumed the clouds to be optically thin nebulae photoionized by ultraviolet radiation (Bahcall and Kozlovsky 1969; Tarter, Tucker, and Salpeter 1969; Davidson 1972, 1975; MacAlpine 1972). *HEAO 1* and *HEAO 2* observations (Tananbaum *et al.* 1979; Zamorani *et al.* 1981; Maccacaro *et al.* 1982) and theoretical studies (Krolik and McKee 1978; Kwan and Krolik 1981) suggest that X-rays create an extended, partially ionized zone in much thicker clouds, with hydrogen column densities $N_{\text{H}} \approx 10^{23} \text{ cm}^{-2}$. State-of-the-art models (Kwan and Krolik 1981; Weisheit, Shields, and Tarter 1981; Ferland and Mushotzky 1984; Kallman 1984) now include sophisticated treatments of the radiative transfer of resonance lines and X-ray ionization processes.

However, some of these models either use approximate values for X-ray secondary heating and ionization, or they omit secondary ionization entirely. This procedure can be a poor approximation. For example, in a medium with ionization fraction $x = n(\text{H}^+)/n(\text{H}_{\text{tot}}) = 0.1$, a single primary electron of energy $E_0 = 2 \text{ keV}$ creates over 24 secondary electrons (17% of its initial energy), deposits 64% of its energy as heat, and leaves the remainder as H I excitation I (15%), and He I excitations or ionizations (4%).

In this paper we describe extensive Monte Carlo calculations which incorporate the best available atomic cross sections for ionization, excitation, and electron scattering. We also provide analytic expressions for the heating, ionization, and H/He line excitation valid over a wide range in both E_0

and x , and we discuss how X-ray secondary heating and ionization may affect models of broad-line clouds.

II. THE CALCULATIONS

Our computations follow the Monte Carlo method described in Paper I (Shull 1979), taking into account the discrete nature of the gas interactions. Bergeron and Collin-Souffrin (1973) have developed an integral formulation of the problem that is valid in the continuous limit. For reasons discussed in the Appendix, we feel that the importance of the discrete nature of the secondary processes following collisional ionization of H and He has been underestimated in the past.

The fractions of a primary electron's energy deposited as heat, secondary ionization, and H/He excitation depend on the ionization fraction x . These processes depend on x , rather than n_e , because the primary's interactions are determined by the relative densities of n_e , $n(\text{H}^0)$, etc. We simulate the interaction of the primary electron with the ambient medium (H I, He I, He II, and electrons) by many Monte Carlo experiments in which the computer follows the degradation of the primary's energy through various interactions with these constituents. Because heavy elements have a total number abundance less than 10^{-2} relative to hydrogen, we neglect them in the simulation. Heavy elements might become important in two circumstances: elevated abundances (e.g., supernova remnants) or situations where $(1 - x) \lesssim 10^{-2}$ for solar abundances.

We consider three generic types of interaction of the primary electron with the gas: (1) collisional ionization of H I, He I, or He II; (2) collisional excitation of H I, He I, or He II; and (3) Coulomb collisions with thermal electrons. The ionizations create additional “secondary” electrons, whose fate must be determined according to tables saved from previous calculations at lower energy. Excitations of He I and II generate line photons (e.g., He I $\lambda 584$, $\lambda 626$; He II $\lambda 304$) which interact further with the gas (see Paper I). The ionization contributions from He amount to less than 5% of the direct ionization by secondaries. However, we have kept separate account of the H and He excitations, so that these rates can be incorporated directly into broad-line models. The ionizations listed below refer only to the direct secondary collisional ionizations of H and He and do not include additional ionizations by He I or II photons.

¹ Also Department of Astrophysical, Planetary, and Atmospheric Sciences, University of Colorado.

Electron-electron collisions transfer heat directly to the gas. Once a primary electron is degraded to energies below 10.2 eV (the Ly α threshold), its energy is deposited entirely as heat, assuming that interactions with molecules or trace heavy elements may be neglected. At high density, Ly α may be collisionally quenched, resulting in additional heating. Here, as with the He lines, we keep separate account of the Ly α excitations, allowing the user to follow their fate directly. For more details about the method, the physical processes, or the assumed cross sections, we refer the interested reader to Paper I.

As in Paper I, our results are based on 1000 Monte Carlo experiments for each primary energy E_0 . Each experiment consists of tracking a large number of energy steps from energy E_0 down to 0. At each step, a random number generator chooses among the processes above; in practice, we choose among seven possibilities corresponding to direct heating by electron-electron collisions, and collisional ionization or excitation of H I, He I, or He II. The probabilities for each process are computed from the relative abundances, $n(e)$, $n(\text{H I})$, $n(\text{He I})$, and $n(\text{He II})$, and from the appropriate atomic cross sections as functions of energy (listed in Paper I). We assume that $n(\text{He})/n(\text{H}) = 0.1$, and that the hydrogen and helium ionization fractions are equal: $n(\text{H}^+)/n(\text{H}_{\text{tot}}) = n(\text{He}^+)/n(\text{He}_{\text{tot}})$. In practice, interactions with He II were never important.

The calculations are also insensitive to the total hydrogen density, which enters only through the Coulomb logarithm in the electron scattering cross section (see Appendix). At very high densities, excitation or ionization from excited states of H or He would become important. However, these effects are negligible in broad-line clouds, where the fractional population in the $n = 2$ levels almost never exceeds 10^{-3} . A thermalized H($n = 2$) population at 10^4 K can exceed n_e when $x \leq 2 \times 10^{-5}$.

As mentioned above, an important ingredient in the calculations of secondary electron effects is a “lookup table” of the heating, ionizations, and excitations produced by the electron released by collisional ionization. For a given ionization fraction, the distribution of a primary electron’s energy among

these forms remains approximately constant for $E_0 > 100$ eV. This scaling with E_0 results from the way in which the primary’s energy is degraded into lower energy secondaries (see the Appendix). In order to automate our calculations, we generate the lookup table for lower energies simultaneously with the calculations. For energies $0 < E_0 < 35$ eV, we proceed in steps of 1 eV; from 35 to 200 eV, in steps of 5 eV; from 200 to 1000 eV, in steps of 25 eV; and from 1000 to 3000 eV, in steps of 100 eV. Beyond 3 keV the rates may be found by extrapolation of our analytic fits.

III. RESULTS

We have run 18 models, with values of ionization x ranging between 0.0001 and 0.95 (Table 1). In each model, we computed the secondary heating, ionizations, and excitations for primary electrons up to 3 keV. Figure 1 shows some of the results in histogram form. For a fixed energy $E_0 = 2$ keV, the heating, ionizations, and excitations are intrinsically distributed over a wide range. The “spikes” in Figure 1 result from discrete excitation or ionization energy losses in multiples of 10.2 and 13.6 eV (H I) or 19.95 and 24.58 eV (He I). This discreteness would be visible in all such panels in Figure 2, except for the expanded scale of H I in the $x = 0.01$ case. Hereafter, we will quote only the means and 1σ of these distributions.

Figure 1 shows the mean amounts of a primary electron’s energy deposited in the forms of heat, ionization, and excitation for selected values of x . Considerable structure below $E_0 = 50$ eV is not apparent on this scale (see Paper I). Because of the code’s accounting scheme, the mean *number* of secondary ionizations or excitations may be evaluated by dividing by the appropriate excitation or ionization energy (10.2 eV, 13.6 eV, etc.). These numbers differ by 10%–20% from those in Paper I, primarily because of an improved grid of cross sections for H I ionization. In Figure 3, we plot the fractional energy deposited as heat and H ionization versus the ionization fraction x for selected values of E_0 . Since E_0 exceeds 100 eV, the curves approach a constant locus, which reflects the linearity of these secondary processes with E_0 at high energy.

TABLE 1
FRACTIONAL ENERGY DEPOSITION^a

MODEL	x	Heat	$\phi(\text{H I})$	$\phi(\text{He I})$	$E(\text{H I})$	$E(\text{He I})$
1.....	0.95	0.9895 ± 0.0108	0.0021 ± 0.0031	0.0004 ± 0.0017	0.0018 ± 0.0025	0.0002 ± 0.0011
2.....	0.90	0.9847 ± 0.0123	0.0041 ± 0.0044	0.0007 ± 0.0025	0.0038 ± 0.0037	0.0003 ± 0.0013
3.....	0.80	0.9726 ± 0.0138	0.0092 ± 0.0064	0.0018 ± 0.0037	0.0090 ± 0.0059	0.0007 ± 0.0022
4.....	0.70	0.9592 ± 0.0161	0.0157 ± 0.0085	0.0029 ± 0.0050	0.0146 ± 0.0071	0.0010 ± 0.0026
5.....	0.50	0.9196 ± 0.0211	0.0345 ± 0.0127	0.0059 ± 0.0069	0.0312 ± 0.0109	0.0023 ± 0.0040
6.....	0.30	0.8459 ± 0.0291	0.0670 ± 0.0180	0.0122 ± 0.0099	0.0629 ± 0.0164	0.0046 ± 0.0055
7.....	0.20	0.7784 ± 0.0355	0.0987 ± 0.0200	0.0173 ± 0.0109	0.0912 ± 0.0212	0.0064 ± 0.0065
8.....	0.15	0.7247 ± 0.0401	0.1223 ± 0.0233	0.0214 ± 0.0129	0.1144 ± 0.0228	0.0085 ± 0.0075
9.....	0.10	0.6524 ± 0.0421	0.1554 ± 0.0244	0.0268 ± 0.0141	0.1458 ± 0.0260	0.0105 ± 0.0082
10.....	0.05	0.5296 ± 0.0426	0.2123 ± 0.0253	0.0351 ± 0.0159	0.1982 ± 0.0280	0.0144 ± 0.0094
11.....	0.02	0.4015 ± 0.0369	0.2683 ± 0.0247	0.0449 ± 0.0174	0.2561 ± 0.0291	0.0179 ± 0.0100
12.....	0.01	0.3256 ± 0.0297	0.3024 ± 0.0238	0.0467 ± 0.0170	0.2943 ± 0.0284	0.0193 ± 0.0100
13.....	0.005	0.2696 ± 0.0249	0.3244 ± 0.0220	0.0499 ± 0.0172	0.3235 ± 0.0281	0.0206 ± 0.0110
14.....	0.002	0.2148 ± 0.0166	0.3452 ± 0.0217	0.0508 ± 0.0179	0.3553 ± 0.0263	0.0219 ± 0.0107
15.....	0.001	0.1852 ± 0.0140	0.3553 ± 0.0209	0.0513 ± 0.0177	0.3736 ± 0.0262	0.0221 ± 0.0112
16.....	0.0005	0.1642 ± 0.0120	0.3593 ± 0.0215	0.0518 ± 0.0178	0.3900 ± 0.0258	0.0224 ± 0.0110
17.....	0.0002	0.1441 ± 0.0098	0.3626 ± 0.0218	0.0523 ± 0.0178	0.4055 ± 0.0244	0.0231 ± 0.0110
18.....	0.0001	0.1358 ± 0.0088	0.3656 ± 0.0217	0.0516 ± 0.0177	0.4119 ± 0.0252	0.0229 ± 0.0108

^a Fraction of a 3 keV primary electron’s energy deposited as heat, secondary ionizations ϕ , and secondary excitations E of H I or He I for 18 values of the background ionization fraction $x = n(\text{H}^+)/n(\text{H}_{\text{tot}})$. The $\pm$ “errors” represent the 1σ widths of the intrinsic distributions (Fig. 1).

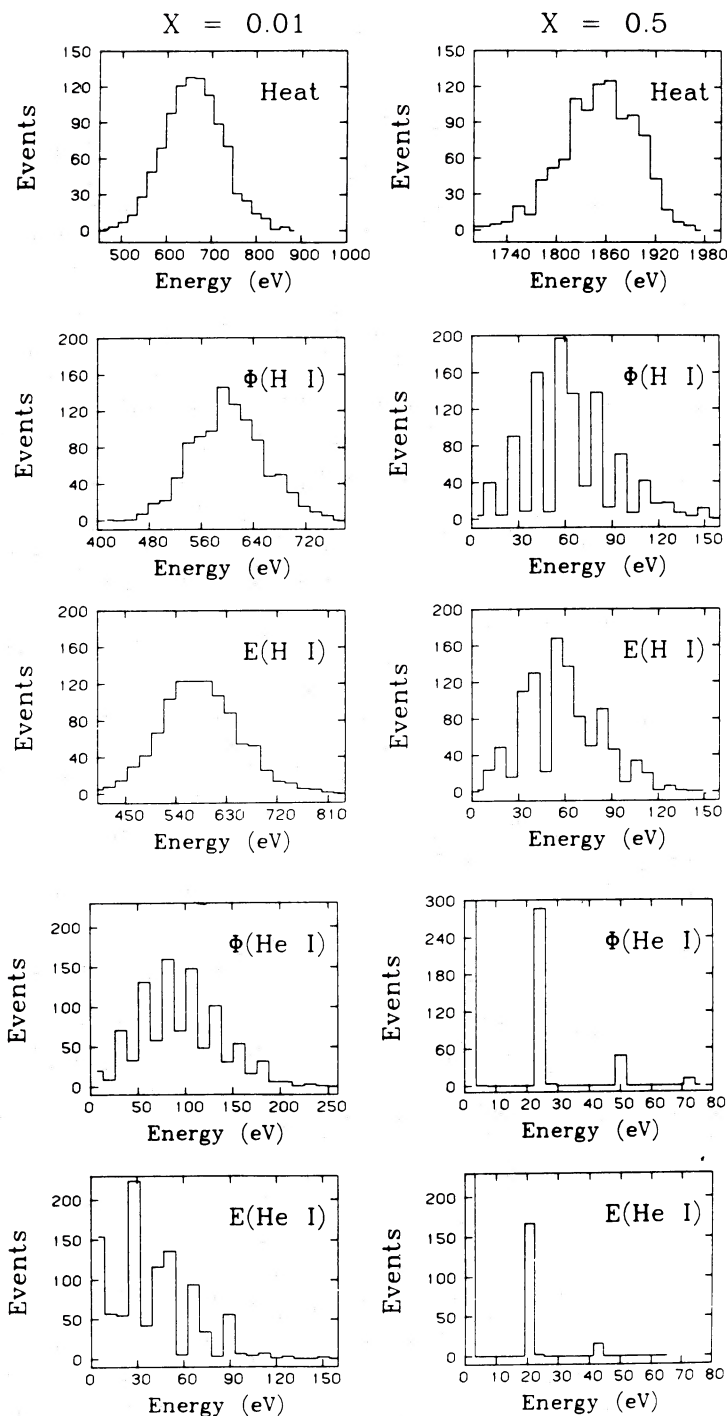


FIG. 1.—Histograms for 1000 trials showing the amount of energy lost by a 2 keV primary electron in the forms of heat, ionizations ϕ , and excitations E of H I or He I for two ionization fractions $x = 0.001$ and $x = 0.5$.

We have checked the accuracy of our Monte Carlo simulations by noting that the sum of all such curves conserves energy to better than 1%. The counting statistics of the Monte Carlo experiments are also of this order.

The limiting values of the fractional energy deposited as heat, ionization, and excitation are shown in Figure 4. Evidently, as x rises above 0.1, a large fraction of the primary's energy goes into heat via Coulomb collisions with the thermal

electrons. However, the detailed manner in which the heating efficiency approaches 100% as $x \rightarrow 1$ is a critical problem in obtaining accurate models of broad-line clouds near the ionization front. These rates are also important at low ionization fractions, deep in the cloud.

Large-scale computer models of X-ray heated nebulae or broad-line clouds require rates for secondary effects over a continuous range in x . We have therefore χ^2 fitted the limiting

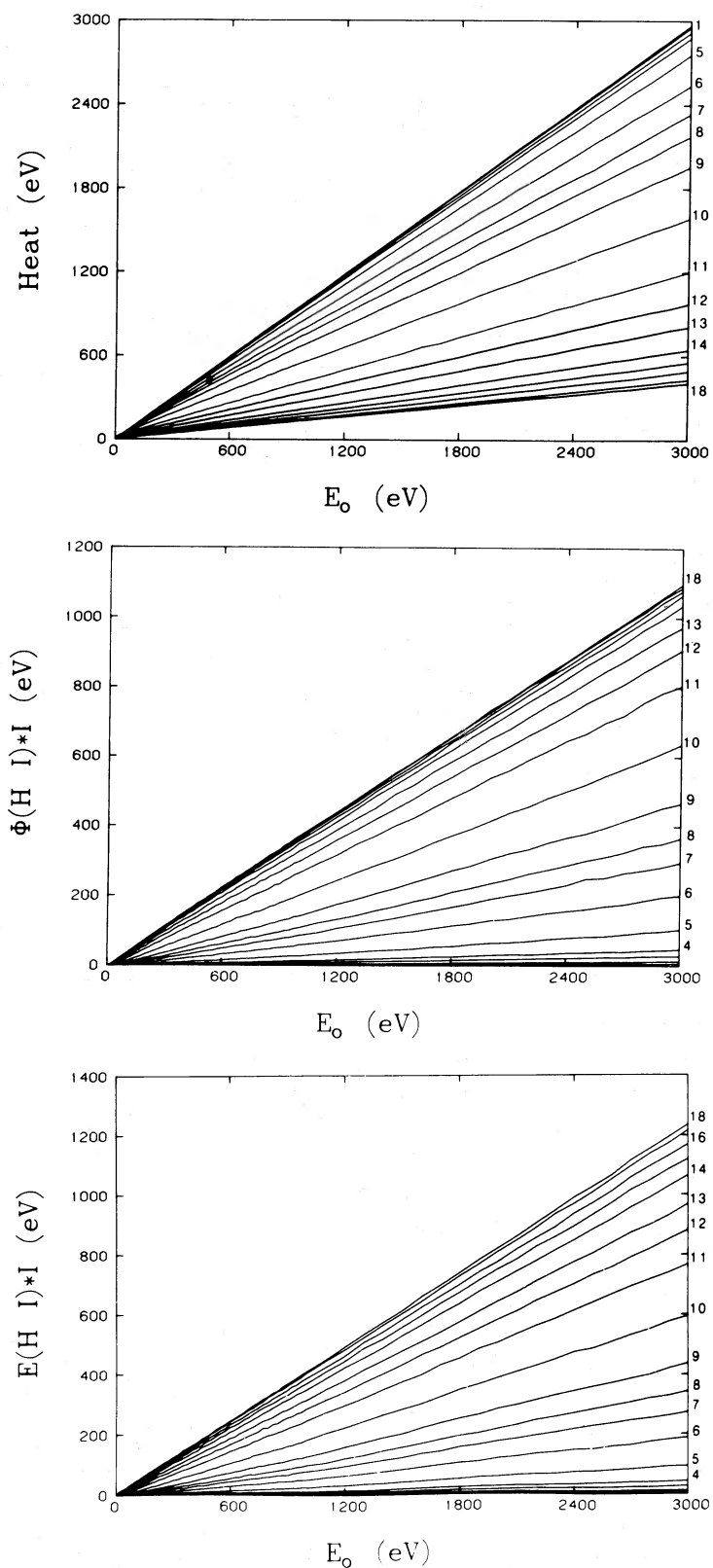


FIG. 2.—Amounts of a primary electron's energy (E_0) deposited as heat, ionization ϕ , and excitation E of H I for 18 values of x (model numbers of Table 1). Here I represents appropriate energy of excitation (10.2 eV) or ionization (13.6 eV).

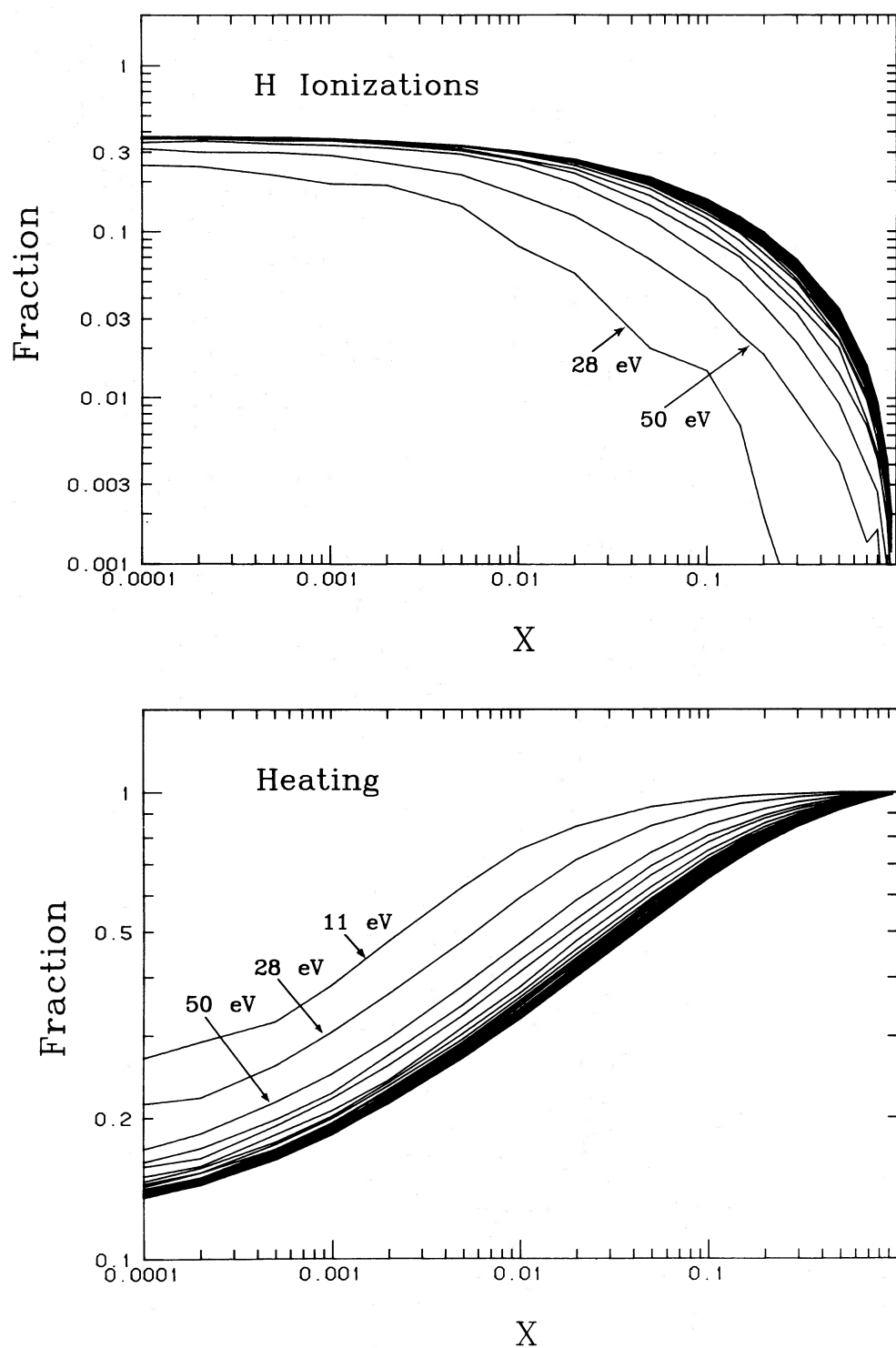


FIG. 3.—Fractional energy deposited as heat and ionization vs. background ionization fraction x for selected values of primary electron energy E_0 . Lowest energies are labeled; curves after that are for $E_0 = 100, 150, 200, 300$ eV, and steps of 100 eV. For $E_0 \gg 100$ eV, curves approach a constant locus.

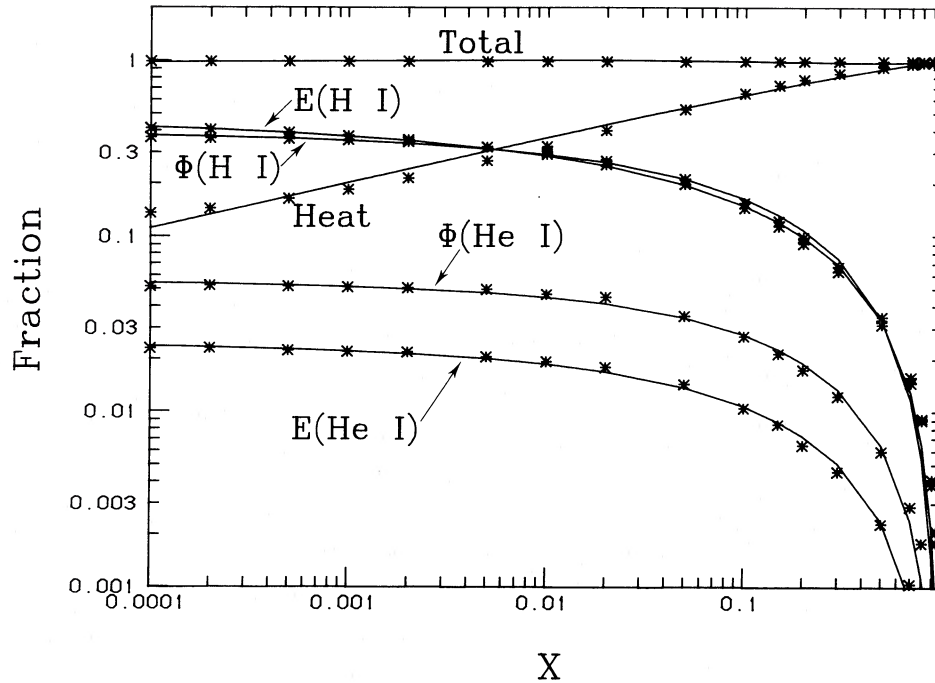


FIG. 4.—Limiting values ($E_0 \gg 100$ eV) for the fraction of a primary's energy deposited as heat, ionization, and excitation vs. ionization fraction x . Solid curves are analytic fits to these values (Table 2).

curves in Figure 3 ($E_0 \gg 100$ eV) to analytic expressions of the following form:

$$\text{Form 1: } y = C[1 - (1 - x^a)^b], \quad (1)$$

$$\text{Form 2: } y = C(1 - x^a)^b. \quad (2)$$

Form 1 was used for the fraction of E_0 deposited as heat; form 2 for the fractional energies going into H and He excitation or ionization. These expressions were chosen because of their appropriate limits at $x = 0$ and $x = 1$, and because their three free parameters allow us to fit the limiting curves to 1%–2% accuracy. Table 2 lists the three constants (C , a , b) for five processes involving heating, and ionization or excitation of H I and He I. Excitations or ionizations of He II were negligibly small, and statistical fluctuations made the fitting coefficients rather uncertain.

Figure 4 also shows the analytic fits to the limiting curves for heating, ionization, and excitation, together with their sum, which reflects energy conservation and the fitting accuracy. Most of the inaccuracy comes from the fit to the heating curve, which is reproduced to $\sim 2\%$ over the range $0.0001 < x < 1.0$.

TABLE 2
FITTING COEFFICIENTS^a

Form	Parameter	C	a	b
1.....	Heat/ E_0	0.9971	0.2663	1.3163
2.....	ϕ (H I) (I/E_0)	0.3908	0.4092	1.7592
2.....	ϕ (He I) (I/E_0)	0.0554	0.4614	1.6660
2.....	E (H I) (I/E_0)	0.4766	0.2735	1.5221
2.....	E (He I) (I/E_0)	0.0246	0.4049	1.6594

^a Coefficients for analytic fits (eqs. [1] and [2]) to heat, secondary ionizations ϕ , and secondary excitations E of H I or He I. Here, I is the appropriate excitation or ionization energy: 10.2 or 13.6 eV for H I; 19.95 or 24.58 eV for He I. These formulae are only accurate for $E_0 > 100$ eV (see Fig. 3).

Our fractions differ considerably from those of Bergeron and Collin-Souffrin (1973). At ionization $x = 10^{-2}$, our fractional heating is 30% higher and our secondary ionization and excitation are 30% lower. The discrepancies in secondary ionization and excitation are considerably higher for $x > 0.5$. These differences presumably reflect our inclusion of helium, our different assumed cross sections, and our treatment of the energy distribution of secondaries following collisional ionization of H and He.

IV. DISCUSSION

X-ray secondary ionization and excitation have often been neglected in AGN photoionization models, although Weisheit, Shields, and Tarter (1981) did include their effects. We have shown that this neglect can be a poor assumption for lines formed at low ionization fractions, particularly for AGNs whose X-ray continua extend to many keV. Because of their penetrating nature, X-rays create an extensive, partially ionized region in which substantial fractions of an X-ray photoelectron's energy can be deposited in collisional ionization or excitation of H and He.

Lines sensitive to temperature or electron density can therefore be affected by secondaries. In addition, Ly α and many other emission lines arise in these partially ionized zones. Photoionization models of broad-line and narrow-line clouds must include these effects when calculating the energy budget and the electron density. For example, [O I], [O II], and [S II] in narrow-line clouds have low critical densities for de-excitation, while [O III] and [Ne III] are temperature sensitive. In broad-line clouds, lines of Mg II, Fe II, and C II $\lambda 2326$ may be affected by changes in both n_e and T . In collaboration with Drs. G. Ferland and T. Kallman, we are examining these predictions using accurate photoionization codes (e.g., Ferland and Mushotzky 1984; Kallman 1984).

We have seen that the secondary electrons deposit their energy almost entirely with H, He, and thermal electrons.

However, for a trace heavy element, the secondaries can provide a significant source of ionization. A detailed computation of the secondary ionization rates of heavy elements depends on the energy spectrum of primaries, and is thus model-dependent. However, to lowest order, one may estimate a trace element's secondary ionization rate by multiplying the H I secondary ionization rate by an appropriate ratio, $\sigma_i(\text{metal})/\sigma_i(\text{H I})$, of collisional ionization cross sections (Lotz 1968). A correction for the different ionization thresholds E_T

would also depend on the primary energy spectrum, but is of order (E_T/E_{primary}) for energetic X-rays.

We thank Drs. Gary Ferland, Tim Kallman, and Julian Krolik for stressing the importance of these calculations for photoionization models and for helpful discussions of the results. This work was supported by NSF grant AST 82-16481 and NASA grant NSG-7128 at the University of Colorado.

APPENDIX

ELECTRON ENERGY LOSSES

The energy degradation of a primary electron in a partially ionized gas is generally understood as a competition between Coulomb collisions with thermal electrons and collision excitation/ionization of neutrals (H I, He I). As the ionization fraction rises, more of the primary's energy goes into heat and less into excitations and ionizations.

However, one aspect of collisional ionization is important for understanding energy degradation of high-energy ($E \gg 100$ eV) electrons. When an energetic electron excites H I, He I, or He II, it loses a small amount of energy, typically 10.2, 20, or 40.8 eV. However, in collisional ionization, the electron typically loses a large amount, owing to the high-energy tail in the energy distribution of secondary electrons (Opal, Peterson, and Beatty 1971; Shull 1979). The average energy of these secondaries is much larger than the values of 8 to 14 eV assumed for much lower energy primaries (Spitzer and Scott 1969; Habing and Goldsmith 1971).

We now discuss the energy dependence of the loss processes. At high energies, the cross sections for Coulomb collisions with electrons (σ_{ee}) and for excitation of H I (σ_{eH}) take the asymptotic (or Born approximation) forms,

$$\sigma_{ee} = (7.82 \times 10^{-11})(0.05/f)(\ln \Lambda)E^{-2} \text{ cm}^2, \quad (\text{A1})$$

$$\sigma_{eH} = (2.75 \times 10^{-15}) \ln(E/13.6 \text{ eV})/E \text{ cm}^2, \quad (\text{A2})$$

where $\ln \Lambda$ is the Coulomb logarithm (Spitzer 1962), E is measured in electron volts, and $f = \Delta E/E = 0.05$ is chosen to simulate the discrete nature of the Coulomb collisions in Monte Carlo models (Habing and Goldsmith 1971; Shull 1979). The collisional ionization cross section is ~ 0.81 times σ_{eH} . Because of the energy dependences of σ_{eH} and σ_{ee} , collisional excitation and ionization of neutral hydrogen dominate the interaction at high energies, when $n(\text{H I})\sigma_{eH} > n(e)\sigma_{ee}$. Numerically this translates into the condition,

$$E \ln(E/13.6) > (2.84 \times 10^4)(0.05/f)[x/(1-x)]. \quad (\text{A3})$$

For example, if $x = 0.1$, then interactions with H I dominate for $E > 780$ eV; if $x = 0.05$ neutral interactions become important for $E > 430$ eV.

In view of this energy dependence, one might expect high-energy primaries to lose most of their energy in ionizations and excitations rather than collisions with electrons. This is not the case, since collisional ionizations of H I (which have an asymptotic cross section $\sim 81\%$ of that for collisional excitation) result in a large energy loss for the primary. Even though the primary electron can collisionally excite H I (losing 10.2 eV), collisional ionizations can divide the energy of one electron among two electrons of lower energy, which then lose a greater portion of their energy in electron collisions. Thus, the discrete nature of the large energy losses in collisional ionization of H I cannot be neglected.

REFERENCES

- Bahcall, J. N., and Kozlovsky, B.-Z. 1969, *Ap. J.*, **155**, 1077.
 Bergeron, J., and Collin-Souffrin, S. 1973, *Astr. Ap.*, **25**, 1.
 Davidson, K. 1972, *Ap. J.*, **171**, 213.
 ———. 1975, *Ap. J.*, **195**, 285.
 Ferland, G. J., and Mushotzky, R. 1984, *Ap. J.*, **286**, 42.
 Habing, H. J., and Goldsmith, D. W. 1971, *Ap. J.*, **166**, 525.
 Kallman, T. 1984, *Ap. J.*, **280**, 269.
 Krolik, J. H., and McKee, C. F. 1978, *Ap. J. Suppl.*, **37**, 459.
 Kwan, J., and Krolik, J. H. 1981, *Ap. J.*, **250**, 478.
 Lotz, W. 1968, *Ap. J. Suppl.*, **14**, 207.
 MacAlpine, G. M. 1972, *Ap. J.*, **175**, 11.
 Maccacaro, T., et al. 1982, *Ap. J.*, **253**, 504.
 Opal, C. B., Peterson, W. H., and Beatty, E. C. 1971, *J. Chem. Phys.*, **55**, 4100.
 Osterbrock, D. E. 1984, *Quart. J. R. A. S.*, **25**, 1.
 Shull, J. M. 1979, *Ap. J.*, **234**, 761.
 Spitzer, L. 1962, *Physics of Fully Ionized Gases* (2d ed; New York: Interscience).
 Spitzer, L., and Scott, E. H. 1969, *Ap. J.*, **158**, 161.
 Tananbaum, H., et al. 1979, *Ap. J. (Letters)*, **234**, L9.
 Tarter, C. B., Tucker, W., and Salpeter, E. E. 1969, *Ap. J.*, **156**, 943.
 Weisheit, J. C., Shields, G. A., and Tarter, C. B. 1981, *Ap. J.*, **245**, 406.
 Zamorani, G., et al. 1981, *Ap. J.*, **245**, 357.

J. MICHAEL SHULL and MICHAEL E. VAN STEENBERG: Joint Institute for Laboratory Astrophysics, University of Colorado, Box 440, University of Colorado, Boulder, CO 80309